ELM 101 7.HAFTA

Haberleşme : Verici ve alıcı arasında 2 tarafın da anlayabildiği herhangi bir ortamı kullanarak bir bilgiyi aktarmasıdır. Haberleşme sistemleri birden fazladır:

- Simplex: one-way communication (e.g., radio, TV)
- Half-duplex: two-way communication but not simultaneous (e.g., push-to-talk radios)
- Full-duplex: two-way communication (e.g., cellular phones)
  - Frequency-division duplex (FDD)
  - Time-division duplex (TDD): simulated full-duplex

3

Underlying concepts

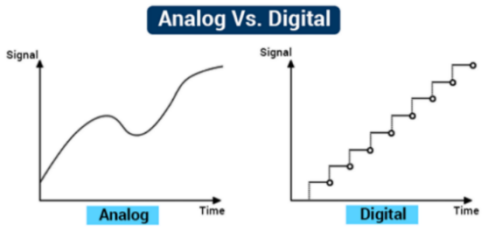
- Electromagnetics
  - Antennas, wave propagation, channel modeling
- Signals and systems
  - Filtering, Fourier transforms, block-diagram design
- Digital signal processing
  - Equalization, spread-spectrum, source coding
- Communications
  - Modulation, noise analysis, channel capacity, channel coding

Enabling technologies

- Digital integrated circuits
- RF generation devices (efficient power amps, sleep modes, improved oscillators, smart antennas)
- Source coding (data compression)
- Modulation (improved efficiency)
- Multiple-access techniques (increase number of users)
- Channel coding/forward error correction (improve probability of successful reception)
- Software programmable radios

Data communication terms

- Data - entities that convey meaning, or information
- Signals - electric or electromagnetic representations of data
- Transmission - communication of data by the propagation and processing of signals



Analog Signals	Digital Signals
Continuous signals	Discrete signals
Represented by sine waves	Represented by square waves
Human voice, natural sound, analog electronic devices are few examples	Computers, optical drives, and other electronic devices
Continuous range of values	Discontinuous values
Records sound waves as they are	Converts into a binary waveform.
Only be used in analog devices	Suited for digital electronics like computers, mobiles and more.

Digital transmission

- Concerned with the content of the signal
- Attenuation destroys integrity of data
- Digital Signal
  - Repeaters achieve greater distance
  - Repeaters recover the signal and retransmit
- Analog signal carrying digital data
  - Retransmission device recovers the digital data from analog signal
  - Generates new, clean analog signal

Analog transmission

- Transmit analog signals without regarding to content
- Attenuation limits length of transmission link
- Cascaded amplifiers boost signal's energy for longer distances but cause distortion
  - Analog data can tolerate distortion
  - Introduces errors in digital data

Classifications of transmission media

- Transmission Medium
  - Physical path between transmitter and receiver
- Guided Media
  - Waves are guided along a solid medium
  - E.g., copper twisted pair, copper coaxial cable, optical fiber
- Unguided Media
  - Provides means of transmission but does not guide electromagnetic signals
  - Usually referred to as wireless transmission
  - E.g., atmosphere, outer space

Unguided media

- Transmission and reception are achieved by means of an antenna
- Configurations for wireless transmission
  - Directional
  - Omnidirectional

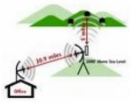
Wired vs. Wireless

Wired



- Low attenuation
- Low time-variance
- Negligible interference
  - Each wire is well isolated
- Costly
- No mobility

Wireless



- Attenuation is high
- Time-varying
- High interference
  - Single medium
- No digging to lay cables
- Nomadic vs. Mobile (?)

- NEDEN WIRELESS HABERLEŞME KULLANIYORUZ ?

Alıcı nerede verici nerede gibi bilgiyi sağlayabiliyor. (Mobility sağlamakta)

Maliyetler düşürülebildi.

- ANTENLER

Temel olarak bir ileticidir. 2 tür:

Transmission: Ortama veriyi yayıyorsa…

Reception: Ortamdan veriyi toplayansa…

2 işlemi de yapan antenler var.

Types of Antennas

- Isotropic antenna (idealized)
  - Radiates power equally in all directions
- Dipole antennas
  - Half-wave dipole antenna (or Hertz antenna)
  - Quarter-wave vertical antenna (or Marconi antenna)
  - Designed to receive particular wavelength
- Parabolic Reflective Antenna

History of wireless comm.

- **Transoceanic Communication**
- 1901 - Marconi successfully transmits radio signal across Atlantic Ocean from Cornwall to Newfoundland
- 1902 - First bidirectional communication across Atlantic
- 1909 - Marconi awarded Nobel prize for physics

- **Voice over Radio**
- 1914 - First voice over radio transmission
- 1920s - Mobile receivers installed in police cars in Detroit
- 1930s - Mobile transmitters developed; radio equipment occupied most of police car trunk
- 1935 - Frequency modulation (FM) demonstrated by Armstrong
- 1940s - Majority of police systems converted to FM

History of wireless comm.

Birth of Mobile Telephony

1946 - First interconnection of mobile users to public switched telephone network (PSTN)

1949 - FCC recognizes mobile radio as new class of service

1940s - Number of mobile users > 50K

1950s - Number of mobile users > 500K

1960s - Number of mobile users > 1.4M

1960s - Improved Mobile Telephone Service (IMTS) introduced; supports full-duplex, auto dial, auto trunking

1976 - Bell Mobile Phone has 543 pay customers using 12 channels in the New York City area; waiting list is 3700 people; service is poor due to blocking

- **Before the "Birth of Radio", 1867-1896**
- 1867 - Maxwell predicts existence of electromagnetic (EM) waves
- 1887 - Hertz proves existence of EM waves; first spark transmitter generates a spark in a receiver several meters away
- 1890 - Branly develops *coherer* for detecting radio waves
- 1896 - **Guglielmo Marconi** demonstrates wireless telegraph to English telegraph office

- **"The Birth of Radio"**
- 1897 - Marconi awarded patent for wireless telegraph
- 1897 - First "Marconi station" established on Needles island to communicate with English coast
- 1898 - Marconi awarded English patent for tuned communication
- 1898 - Wireless telegraphic connection between England and France established



Cellular summary

	1ST GENERATION UBIQUITY	2ND GENERATION DIGITAL	3RD GENERATION DATA
Time	1980s	1990s	2000s
Technology	Analog	Digital (TDMA,CDMA)	Packet
Services	Voice, Data Modems	Voice, Circuit + Packet Data	Multimedia
Systems	AMPS, TACS …	IS-95, GSM, …	W-CDMA, cdma2000, …
Data Rates	10 Kb/s	~ 20 Kb/s	384 Kb/s and higher
Key Issues	*viability*--would cellular work	*coverage and capacity*--would cellular meet demands	*wireless web, location based services*--what data service drives demand

AMPS: Advanced Mobile Phone System
TACS: Total Access Communication System

Mobile and personal communication networks

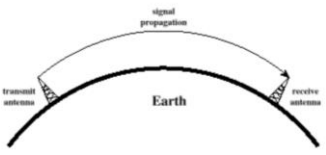
– Bluetooth, – WiFi, – Ad hoc networks, – Sensor networks – ZigBee radios, – UWB, – …

(ANALOGDAN DİJİTALLEŞMEYE SERÜVEN. HIZ VE YÖNTEMLERIN GELIŞMESI INCELENMESI…)

- DALGA ILETIMI IHTIYACI VAR. BU NEDENLE ÇEŞITLI "WAVE PROPAGATION" 3 TIPTE INCELENIR…

Ground Wave Propagation

- Follows contour of the earth
- Can propagate considerable distances
- Frequencies up to 2 MHz
- Example: AM radio

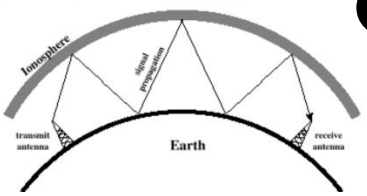


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Sky Wave Propagation

- Signal reflected from ionized layer of atmosphere back down to earth
- Signal can travel a number of hops, back and forth between ionosphere and earth's surface
- Examples: amateur radio, CB radio

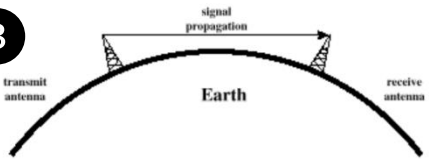
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Line-of-Sight Propagation

- Transmitting and receiving antennas must be within line of sight
  - Satellite communication – signal above 30 MHz not reflected by ionosphere
  - Ground communication – antennas within *effective* line of site due to refraction

3



DALGA TIPLERINDE YAPTIĞI HAREKET TIPINE BAKILABILIR…

SİNYAL BOZUCU DURUMLAR (IMPAIRMENTS):

- *Attenuation*: Sinyal gücündeki "Zayıflama". En çok da uzaklık gibi şeylerle alakalıdır.

- *Free Space Loss*: 1. Maddenin alt dalıdır. Uzaklığın karesi gibi doğrusal olarak düşüyor.

- *Multipath*: alttaki etkiler nedeniyle alıcıda verinin birden fazla kopya olması:
  1) Reflection: (İşaret) Kendinden büyük dalga boyuna çarpmasıdır.
  2) Scattering: (Saçılma) Kendinden küçük dalga boyuna çarpmasıdır.
  3) Diffraction: (Kırılma) Keskin yerlere çarpmasıdır.
  4) Refraction: Ortamdan ortama geçmesidir.

- Gürültü
- Atmosferik emilim nedenleri de mevcut. (örn. sis, çiğ)

OSI Reference Model

- Functionalities of a communication system can be characterized in terms of 7 abstraction layers

